

HCMC UNIVERSITY OF TECHNOLOGY AND ENGINEERING

FACULTY OF ADVANCED EDUCATION



FINAL TERM PROJECT

**ANALYZE THE IMPACT OF PCA ON
MULTISPECTRAL IMAGES**

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Course: Mathematical For Artificial Intelligence

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INTRODUCTION

1. Background:

In the era of modern Earth observation, Sentinel-2 satellite imagery has become a cornerstone for environmental monitoring, land-use classification, and urban planning. These multispectral images, often comprising 14 or more spectral bands, provide a wealth of information but also present significant computational challenges. The high-dimensionality of this data lead to the "Curse of Dimensionality," where the volume of the space increases so rapidly that the available data becomes sparse. Consequently, training machine learning models directly on raw multispectral data demands extensive computational time and high RAM utilization. In an environment where hardware resources are costly and real-time processing is increasingly prioritized, finding an efficient way to condense this data is essential.

2. Motivation:

The motivation for this project stems from the need for resource-efficient AI. Traditional classification pipelines often struggle with the memory footprint required to store and process millions of pixels across multiple spectral bands. By investigating Principal Component Analysis (PCA), we aim to transform the original correlated bands into a set of linearly uncorrelated components. This mathematical transformation is not merely about data reduction; it is about "information distillation"—retaining the most significant variance while discarding noise and redundant features. An optimized pipeline could allow sophisticated models like Support Vector Machines (SVM) to operate on lower-end hardware without a substantial loss in classification performance.

3. Research Objectives:

The primary objective of this study is to evaluate the impact of PCA-based dimensionality reduction on the classification of EuroSAT multispectral images. Specifically, the project aims to:

- Quantify the reduction in RAM consumption and training time when using PCA compared to full-band data.

- Assess the trade-off between the number of principal components retained and the final SVM classification accuracy.
- Develop a robust mathematical pipeline that balances computational economy with high-fidelity results.

4. Scope of Work

This research is conducted using the EuroSAT dataset, consisting of 27,000 multispectral image patches. The technical scope includes:

- Data Pre-processing: Implementing standardization and batch-based covariance matrix computation for over 4 million pixels.
- Linear Algebra Application: Executing Eigen-decomposition to extract the top Principal Components.
- Classification: Training and testing a non-linear SVM with RBF Kernel on the reduced feature space.
- Performance Metrics: Measuring and comparing accuracy, processing time, and memory usage across different compression levels.

RELATED WORK

The field of Land Use and Land Cover (LULC) classification has seen significant advancements with the introduction of the EuroSAT dataset by Helber et al. [1]. Their work established a high-performance benchmark using deep Convolutional Neural Networks (CNNs) on Sentinel-2 multispectral imagery, achieving an accuracy of 98.57%. However, such deep learning approaches often require substantial computational power and memory.

Complementary to this, the challenges regarding the "massive high-dimensional" nature of multispectral data are addressed by Giannopoulos et al. [2]. Their research emphasizes the necessity of resource-efficient schemes to handle communication and storage constraints. They demonstrate that processing compressed or low-rank representations of multispectral data is crucial for practical remote sensing applications.

Building upon these foundations, our project bridges the gap between high-dimensional data processing and resource efficiency. Unlike the complex tensor-based or deep learning methods, we investigate the impact of Principal Component Analysis (PCA) as a classical yet powerful mathematical tool to reduce the feature space. Our work evaluates how this reduction directly affects the trade-off between SVM classification performance and hardware resource consumption (RAM and Time), providing a pragmatic perspective for AI deployment on constrained systems.

CHAPTER 1: THEORETICAL BACKGROUND

1.1. Overview of EuroSAT Dataset

The EuroSAT dataset, introduced by Helber et al. [1], is a breakthrough in providing benchmark data for the Land Use and Land Cover (LULC) classification problem. Data were collected from the Sentinel-2 satellite of the European Copernicus Earth Observatory program.

EuroSAT comprises a total of 27,000 geolabeled and referenced images, categorized into 10 different terrain object classes: Industrial, Residential, Annual Crop, Permanent Crop, River, Sea Lake, Herbaceous Vegetation, Highway, Pasture, and Forest.

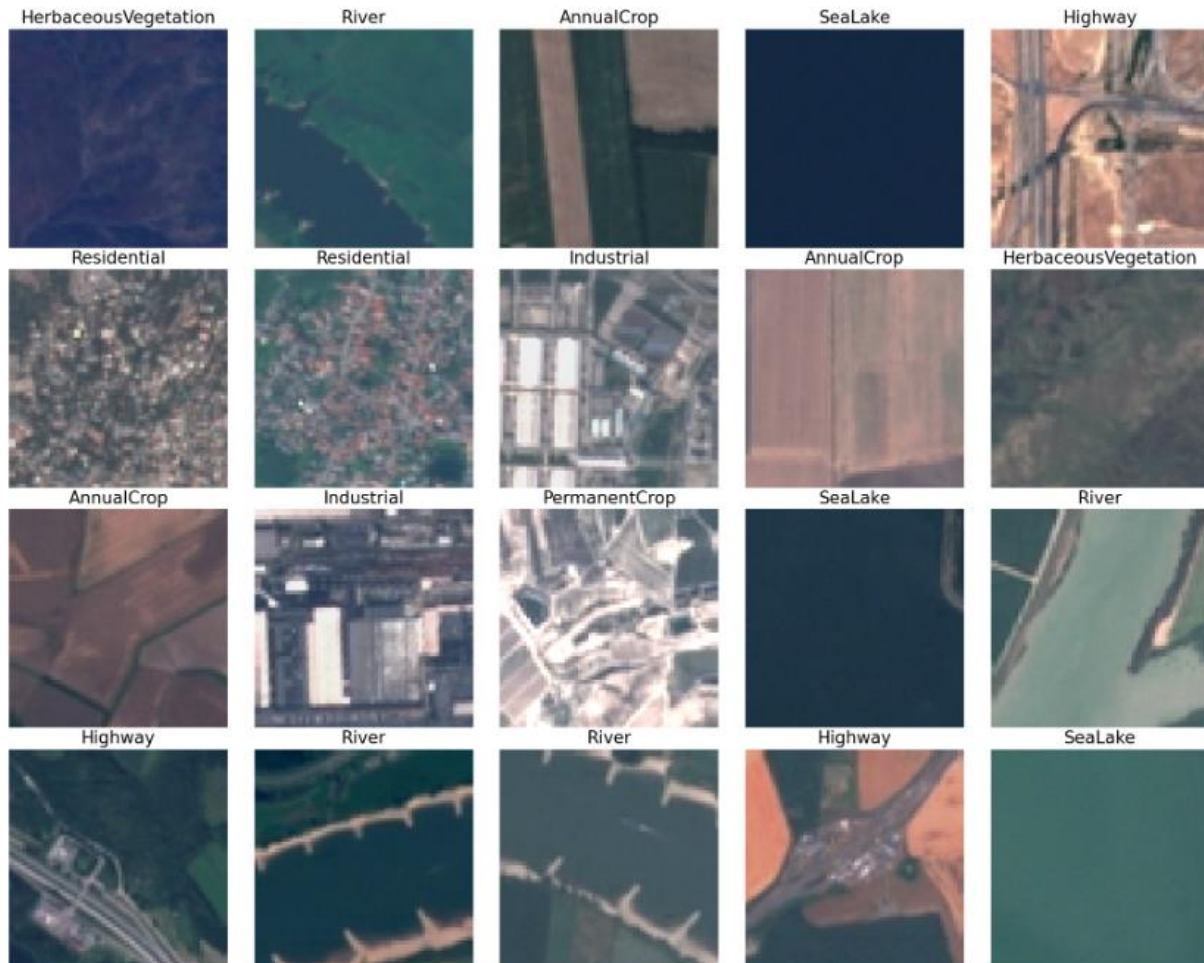


Fig 1. Some samples of EuroSAT images

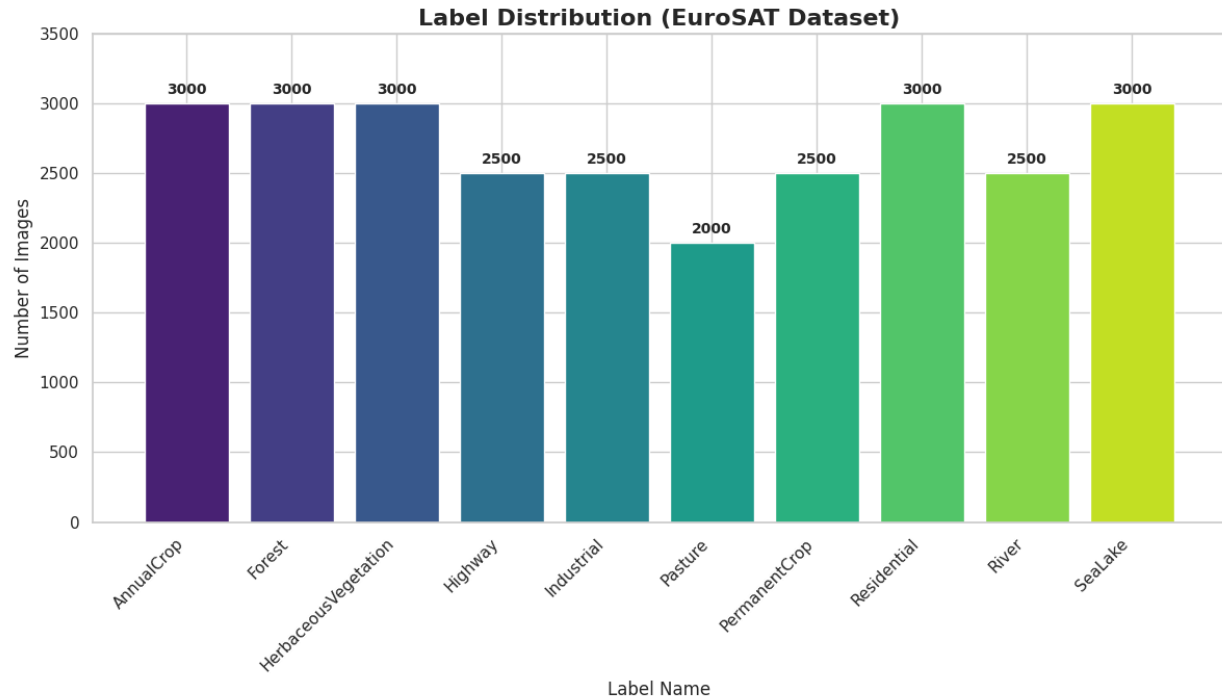


Fig 2. Representation of the image quantity per label in the EuroSAT dataset.

Technically, each image sample is a 64x64 pixel (640m x 640m) data cube with 13 spectral bands. Unlike conventional RGB images which only record visible light intensity, the .tif files in EuroSAT store actual reflectance values at different wavelengths, from blue (Band 2) to shortwave infrared (Band 12).

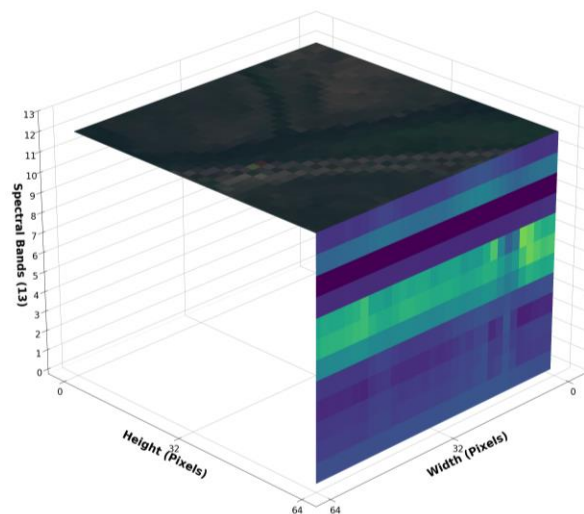


Fig3. Visualization of the EuroSAT multispectral data cube

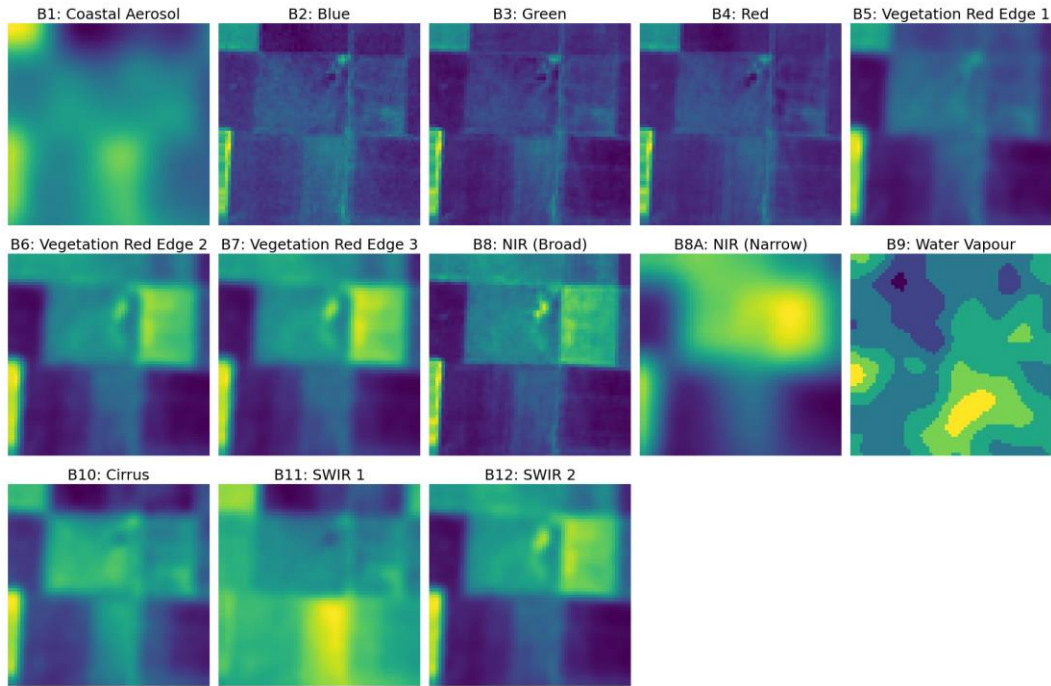


Fig 4. Presenting 13 bands of an images

Each 64x64 image in EuroSAT contains 13 spectral bands. Through experimentation, we found that each of the 10 label layers possesses a unique Spectral Signature.

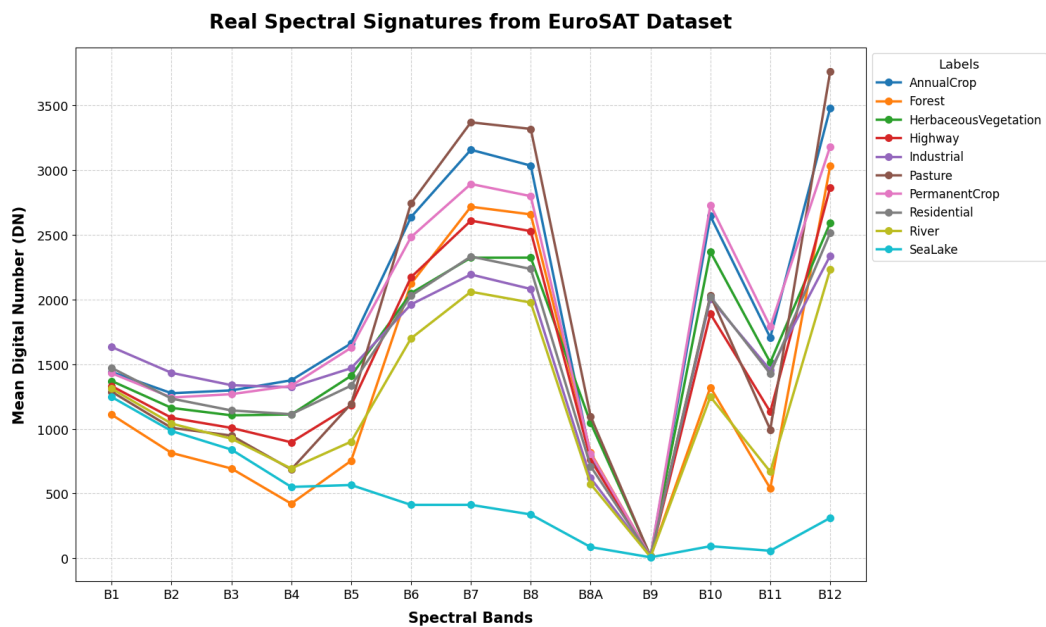


Fig 5. Spectral signature from EuroSAT Dataset

Vegetation tends to reflect strongly in the near-infrared (Bands 8) but absorb strongly in the red (Band 4) range. A key finding from EDA is the significant overlap in value distribution between the Pasture and Forest classes. This suggests that a simple linear classifier would fail, and that compressed features from PCA would be needed to highlight the differences.

Although the spectral signature provides valuable classification features, statistically, the existence of 13 spectral bandscauses multicollinearity. By calculating the correlation matrix between the frequency bands, we found the following groups of spectral bandshave large correlation coefficients, indicating data overlap.



Fig 6. A heatmap shows the correlation between spectral frequency bands

Based on the EDA analyses mentioned above, Principal Component Analysis (PCA) was chosen as the "filter" to extract the most essential information. PCA doesn't simply remove frequency bands; it performs a linear transformation to create new Principal Components (PCs).

1.2. Multispectral Data Representation in Remote Sensing

For machine learning algorithms to harness the power of multispectral data, the transformation from physical entities to digital data structures is essential. However, this process is where the conflict arises between information richness and hardware resource limitations.

Structurally, each EuroSAT image sample is a data cube of size $64 \times 64 \times 13$. To be fed into the SVM classifier, this data cube must undergo a flattening process to become a unique feature vector. Mathematically, each pixel is no longer a point but becomes a point in 13-dimensional space. Thus, a single image is transformed into a vector of length:

$$P = 64 \times 64 \times 13 = 53,248 \text{ features}$$

With a total size of 27,000 samples, the global input matrix would have a size of $27,000 \times 53,248$. In a typical computing environment, storing this matrix as a floating-point number (float64) would consume:

$$\text{Memory} \approx 27,000 \times 53,248 \times 8 \text{ bytes} \approx 11.5 \text{ GB}$$

However, in practical implementation, calculations of the covariance matrix and kernel matrix will require intermediate variables, pushing RAM usage above 30 GB. This is beyond the capabilities of most current personal computer systems, creating a physical barrier to training the model directly on raw data.

Faced with the challenge of memory, this research does not take the conventional approach of processing raw global data. Instead, we designed a "Batch-based Learning" strategy to optimize the data flow:

- **Information Segmentation:** The dataset of 27,000 images was divided into 27 batches. Each batch containing roughly 1,000 samples with first 64×64 rows and 13 columns (features) stand for sample 1 and the same with rests. This method helps limit the amount of data that needs to be processed at any one time, ensuring the system always operates within the safe limits of RAM.
- **Stratified Sampling Integrity:** To preserve data objectivity, each batch is structured using stratified sampling. This ensures that the proportion of the 10 label classes is perfectly balanced in each processing run, preventing model bias due to a lack of samples of any particular label.

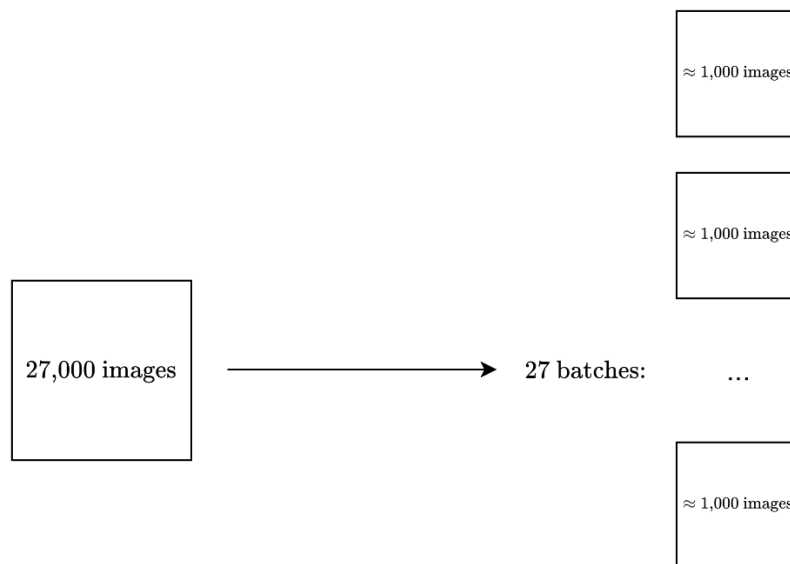


Fig 7. Dividing data into 27 separated batches

This data splitting strategy is not just a programming trick, but a tactical preparation step. It creates a data filtering funnel, where PCA acts as an information extraction engine, compressing this massive amount of data into its most essential components before transferring it to the SVM classifier.

***NDVI:**

In this research, we do not solely rely on the raw spectral values. To improve the model's ability to distinguish between vegetation and non-vegetation classes, we perform Feature Engineering by calculating the Normalized Difference Vegetation Index (NDVI).

Mathematically, NDVI is derived from the contrast between the Red band (Band 4) and the Near-Infrared band (Band 8):

$$NDVI = \frac{NIR - Red}{NIR + Red} = \frac{Band8 - Band4}{Band8 + Band4}$$

In our proposed batch-processing strategy, each 1,000-sample batch undergoes a transformation where a new column is appended to the feature matrix. Consequently, each image's representation is expanded from 13 to 14 features.

1.3. Principal Component Analysis (PCA) for Information Distillation

Principal Component Analysis (PCA) is not simply a data dimensionality reduction technique, but rather acts as an "information distillation" mechanism. In the context of EuroSAT data, PCA performs a linear transformation to map the data from its original high-dimensional space to a new coordinate system where the coordinate axes are independent components and carry the most information.

Before performing geometric transformations, the multispectral data must be brought to a common scale. Each spectral band in the global matrix is standardized using the formula:

$$z = \frac{x - \mu}{\sigma}$$

In satellite imagery, reflectance values across different bands (e.g., Blue vs. SWIR) exhibit significantly different ranges. Without standardization, PCA would be biased

toward bands with larger absolute scales, mistakenly identifying them as having higher variance. This step ensures that every spectral band contributes equally to the feature extraction process, preventing "dominant" bands from overshadowing subtle but critical spectral information.

This step serves as the mathematical bridge from our empirical EDA observations (the Heatmap in Fig 6) to an algebraic structure. The covariance matrix C is calculated as:

$$C = \frac{1}{n-1} X^T X$$

The $d \times d$ matrix C summarizes the correlation between all spectral bands. The off-diagonal elements reflect the information redundancy discovered during EDA. The goal of PCA is to find a new basis where the covariance matrix becomes diagonal, effectively eliminating all multicollinearity and redundant "noise."

The eigenvalues are the roots of the characteristic equation:

$$\det(C - \lambda I) = 0$$

The final stage involves solving the characteristic equation to find the eigenvalues and eigenvectors:

$$Cv = \lambda v$$

- **Eigenvalues (λ):** Quantify the amount of variance (information) captured by each Principal Component (PC). In this study, we sort λ in descending order: $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$.
- **Eigenvectors (v):** Determine the direction of the new axes in the feature space.

By selecting the top k components ($k \leq 14$) such that the cumulative variance ratio reaches a target threshold (e.g., 95%), we condense the massive data cube into a

lean representation. Retaining the largest eigenvalues ensures we keep the "essence" of the land cover characteristics while discarding smaller eigenvalues that typically represent spectral noise.

1.4. Support Vector Machine (SVM) in Reduced Feature Space

Following the information distillation via PCA, a Support Vector Machine (SVM) is employed as the robust core classifier. SVM is a powerful supervised learning algorithm designed to find the optimal decision boundary that separates different classes in a high-dimensional feature space.

The primary objective of the SVM is to identify an Optimal Hyperplane that separates the 10 terrain classes of the EuroSAT dataset with the maximum possible Margin. The margin is defined as the distance between the hyperplane and the nearest data points from each class, known as Support Vectors.

The optimization problem is formulated as follows:

$$\min_{w,b} \underbrace{\frac{1}{2} \|w\|^2}_{\text{Margin Maximization}} + \underbrace{C \sum_{i=1}^n \zeta_i}_{\text{Penalty for Misclassification}}$$

In this equation, the parameter C plays a critical role as a regularization constant. It balances the trade-off between maximizing the margin width and minimizing the classification error on the training data. For multispectral imagery, where subtle atmospheric noise or sensor variations may exist, a well-tuned C value ensures the model maintains high generalization power and avoids overfitting.

As established in the Exploratory Data Analysis (Section 1.1), many LULC classes—such as Forest and Pasture—exhibit significant spectral overlap, making them linearly inseparable in the initial feature space. To overcome this, we utilize the Radial Basis Function (RBF) Kernel:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

The RBF kernel allows the SVM to map the PCA-reduced features into a higher-dimensional implicit space where the complex, non-linear boundaries of terrain objects become linearly separable. While PCA performs physical dimensionality reduction to save computational resources, the RBF kernel performs implicit dimensionality expansion to capture the intricate nuances of the spectral signatures.

CHAPTER 2: DATA PROCESSING PIPELINE

This chapter details the systematic transformation of raw EuroSAT multispectral data into an optimized feature space suitable for Support Vector Machine (SVM) classification. The pipeline is specifically designed to handle massive data dimensions within the constraints of standard hardware memory (RAM).

**Pipeline Architecture Overview:*

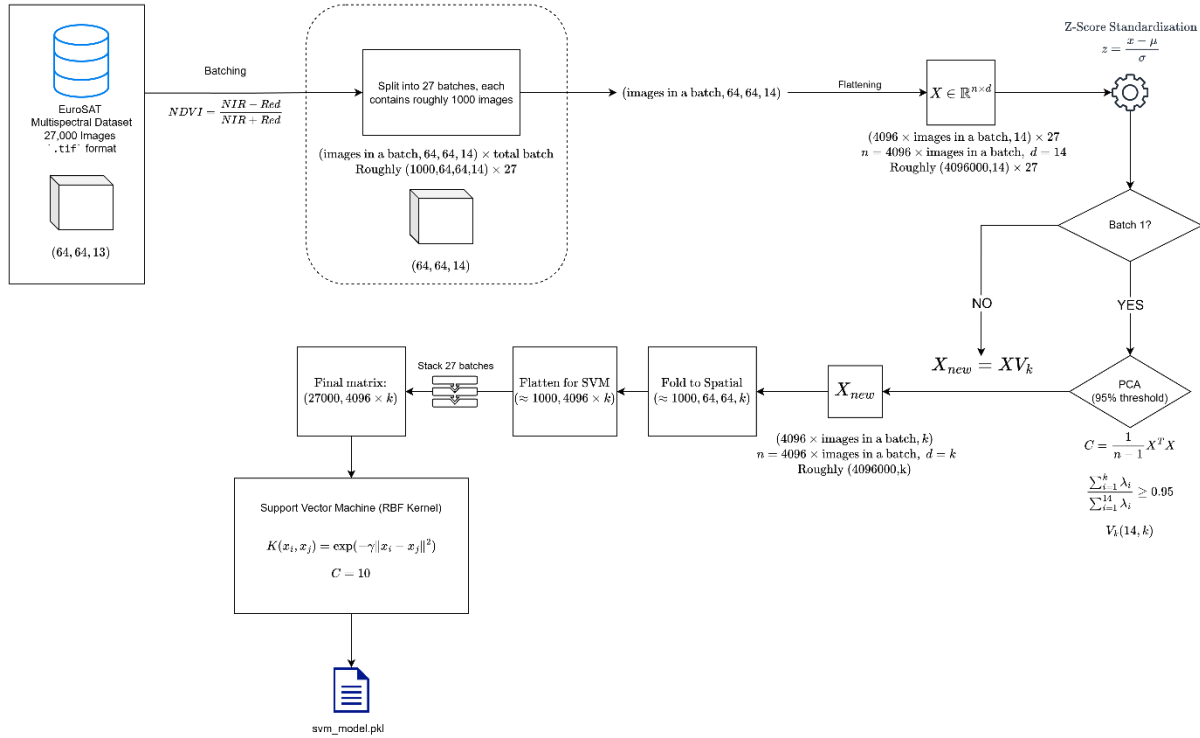


Fig 8. Pipeline Architecture

2.1. Data Preparation and Statistical Normalization

The initial stage of the pipeline focuses on transforming the raw repository of .tif files into a structured, computationally efficient format while addressing the critical challenge of hardware memory constraints. The goal of this stage is to convert raw image files into a structured, standardized numerical format.

2.1.1. Stratified Data Partitioning

To maintain the statistical representativeness of the dataset throughout the learning process, we implemented a Stratified Batching strategy. Unlike simple random sampling, this method ensures that the class distribution within each batch mirrors the global distribution of the EuroSAT dataset (comprising 10 terrain classes).

$$P(y = k|D) \approx P(y = k|B_j)$$

Where D is the whole dataset, B_j is the j^{th} batch and k is the label of the data (0-9). If the data is generated using pure random sampling, it's possible that a batch might not contain any "Forest" images, leading to systematic bias.

Images within each class are shuffled independently to prevent any ordering bias from the original file system. the images from each class are divided into 27 equal parts and distributed across the batches. Each resulting batch contains approximately 1,000 images (ranging from 997 to 1007), providing a consistent workload for subsequent processing stages.

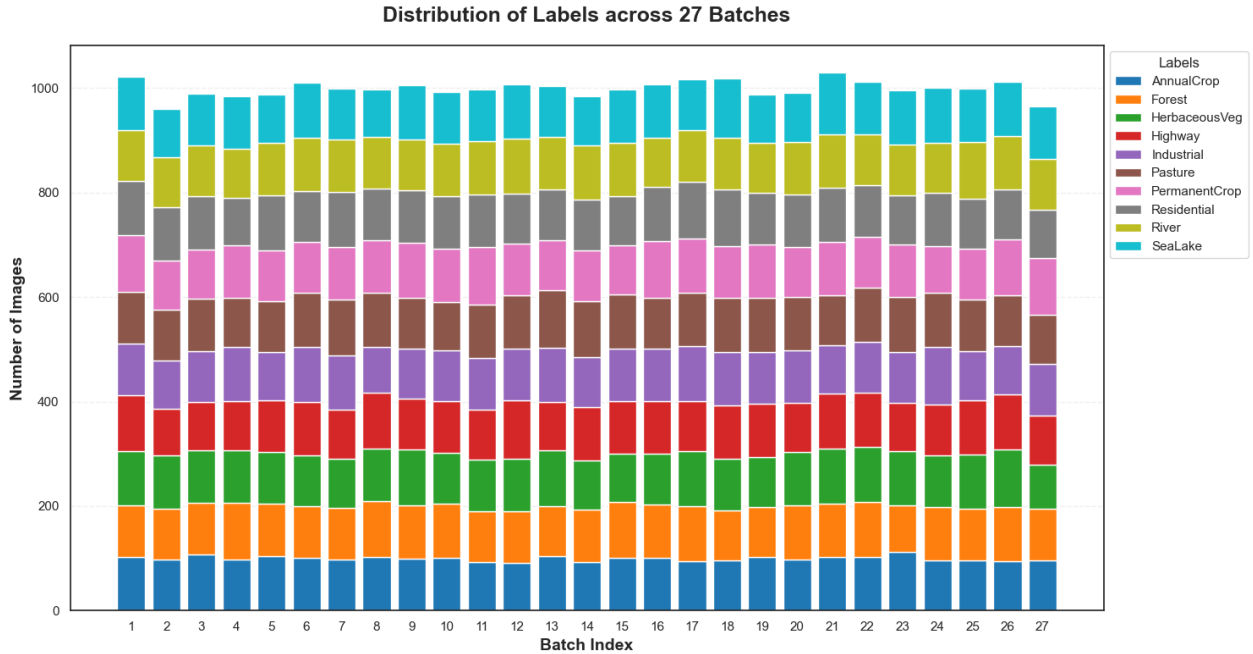


Fig 9. Distribution of labels across 27 batches

2.1.2. Feature Engineering and Pixel-level Flattening

During the ingestion of each batch, every multispectral image undergoes a transformation from a raw spatial cube into an augmented feature matrix.

For every $64 \times 64 \times 13$ cube, we extract the Red (Band 4) and NIR (Band 8) layers to calculate the NDVI. The calculated NDVI layer is concatenated as the 14th band, expanding the data cube to $(14, 64, 64)$

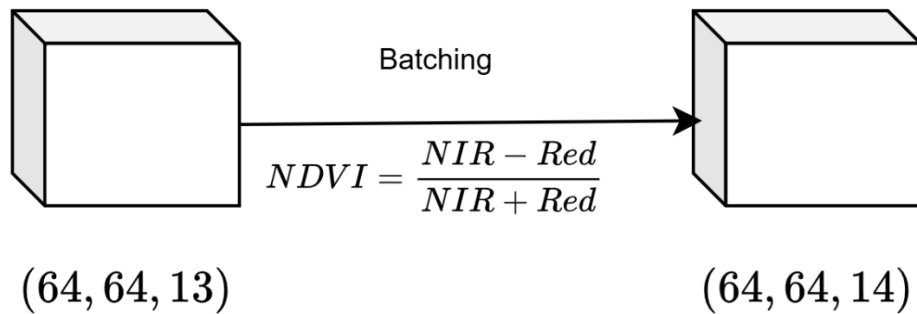


Fig 10. Feature extraction – NDVI

To prepare for pixel-wise statistical analysis, each image is reshaped into a matrix of $(4096, 14)$. In this representation, each of the 4,096 pixels is treated as an individual observation with 14 spectral features.

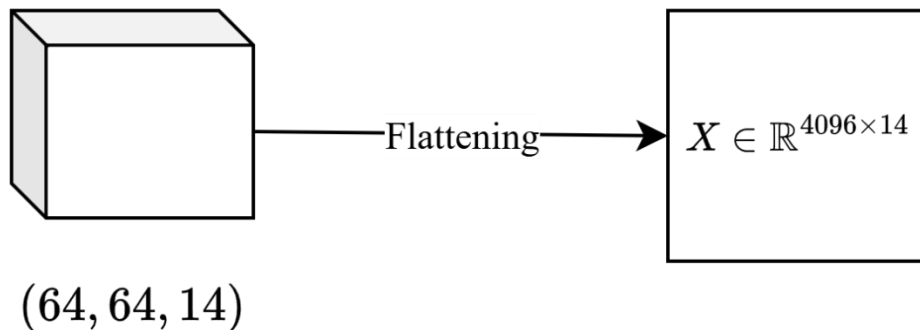


Fig 11. Image flattening

2.1.3. Stacking and Storage

The next step of Stage 1 is the conversion of these processed lists into high-performance NumPy arrays and their serialization to disk. All images within a batch are vertically stacked using `np.vstack`. For a standard batch of 1,000 images, the output matrix X_{batch} reaches a size of $(\approx 4,096,000 \times 14)$. These matrices, along with their corresponding label arrays y_{batch} , are saved as `.npy`

$$X_{batch} = \text{stack}(X_1, X_2, \dots, X_m) = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_m \end{bmatrix}$$

Fig 12. Stacking matrix

$$X \in \mathbb{R}^{n \times d}$$

$$\begin{aligned} & (4096 \times \text{images in a batch}, 14) \times 27 \\ n &= 4096 \times \text{images in a batch}, d = 14 \\ & \text{Roughly } (4096000, 14) \times 27 \end{aligned}$$

Fig 13. Final dimension of each batch

Finally, the whole 27k images have transferred into 27 batches with size of $(\approx 4,096,000 \times 14)$ each batch.

2.2. Dimensionality Reduction via Covariance Matrix and Eigen-decomposition

The second stage of the pipeline involves distilling the high-dimensional standardized data into a lean, optimized feature space. Instead of using black-box reduction

techniques, we implement a manual **Principal Component Analysis (PCA)** approach based on the statistical relationships between spectral bands.

2.2.1. Batch-wise Z-Score Standardization

Before performing dimensionality reduction, the flattened batch matrix must be statistically prepared. Because spectral bands and the engineered NDVI operate on different physical scales, we apply Z-Score Standardization.

For each batch matrix $X_i \in \mathbb{R}^{n \times 14}$, each feature column j is transformed:

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

This ensures that the mean (μ) is 0 and the standard deviation (σ) is 1 for every feature. This step is critical because PCA is sensitive to the variances of the initial variables; without standardization, bands with larger numerical ranges would inappropriately dominate the principal components.

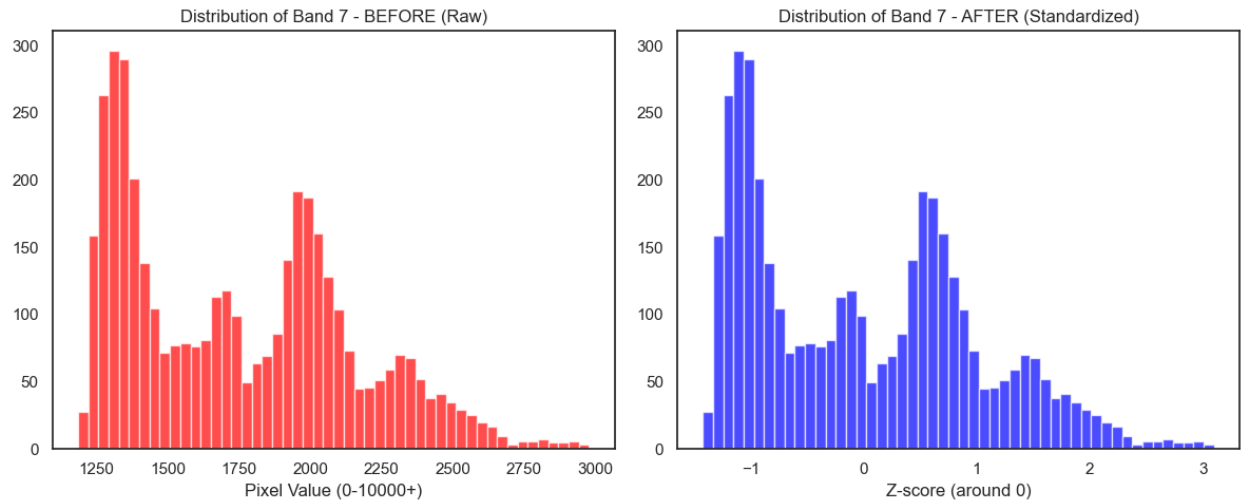


Fig 14. Distribution of Band 7 before and after Z-score

2.2.2. Covariance Matrix Computation

The core of our PCA approach lies in identifying how the 14 spectral features vary together. We compute the Covariance Matrix C from the standardized batch. Using the standardized matrix X , the 14×14 covariance matrix is calculated as:

$$C = \frac{1}{n-1} X_{scaled}^T X_{scaled}$$

The 14×14 matrix C captures how much each spectral band varies in relation to others. The off-diagonal elements represent the redundancy discovered during EDA. Our goal is to transform this matrix into a diagonal form, where all off-diagonal correlations become zero.

To find the axes that capture the maximum variance, we perform Eigen-decomposition. This involves solving the Characteristic Equation:

$$\det(C - \lambda I) = 0$$

Where:

- C : The Covariance Matrix.
- λ : The eigenvalues (scalar values representing the variance of each component).
- I : The identity matrix.

Solving this determinant results in a 14th-degree polynomial, whose roots are the Eigenvalues ($\lambda_1, \lambda_2, \dots, \lambda_{14}$).

For each eigenvalue λ_i , we solve the linear system to find its corresponding Eigenvector v :

$$(C - \lambda I)v = 0$$

These unit vectors define the direction of the new Principal Components in the 14-dimensional space. The eigenvalues, sorted in descending order $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{14}$,

represent the amount of variance captured by each principal component. We determine the optimal number of components k by reaching a 95% cumulative variance threshold:

$$V_{total} = \sum_{i=1}^{14} \lambda_i \longrightarrow R_k = \frac{\sum_{i=1}^k \lambda_i}{V_{total}}$$

This mathematical rule allows us to retain the vast majority of the dataset's "significance" while discarding spectral noise and redundant features. Our experiments lead to $k = 5$.

2.2.3. Dimensional Projection

Finally, we create the projection matrix $V_k \in \mathbb{R}^{14 \times k}$ containing the top k eigenvectors. The standardized data is then projected into the new reduced space:

$$X_{compressed} = X_{scaled} \cdot V_k$$

$$X_{scaled} \in \mathbb{R}^{n \times 14}$$

Now, the compressed matrix $X_{compressed} \in \mathbb{R}^{4096000 \times k}$ where $k \leq 14$.

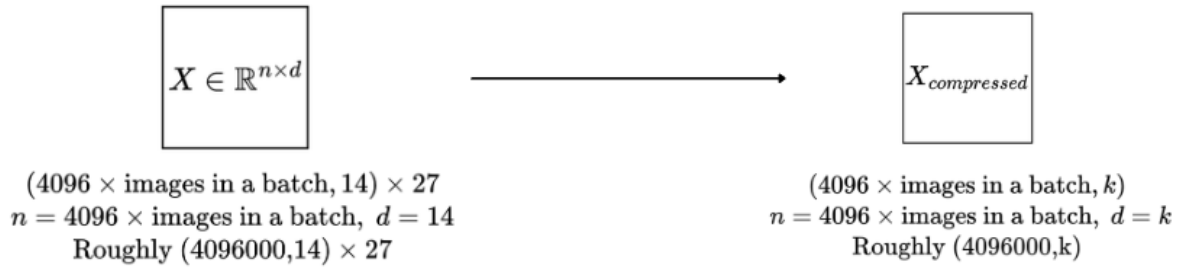


Fig 15. Reduced dimensions matrix transformation

Operation	Input Shape	Output Shape	Objective
Standardization	(n, 14)	(n, 14)	Equalize feature scales.
Covariance	(n, 14)	(14, 14)	Capture inter-band correlations.

Eigen-decomp	(14, 14)	$V_k(14, k)$	Identify optimal projection axes.
Projection	$(n, 14) \cdot (14, k)$	(n, k)	Compress data into k dimensions (with $k = 5$).

Table 1. Summary of Matrix Transformations in Stage 2

2.3. Spatial Reconstruction and Global Feature Integration

The final stage of the data processing pipeline involves reassembling the compressed pixel data into spatial structures and consolidating all batches. This stage ensures that the transformed data maintains its image-level integrity, providing the Support Vector Machine (SVM) with a structured input that combines spectral distillation with spatial context.

2.3.1 Spatial Re-folding

After the PCA projection in Stage 2, the data exists as a continuous matrix of pixel-observations of size $(4,096,000 \times k)$. To perform image-level classification, the pipeline must restore the spatial relationship between these pixels.

Each sequence of 4,096 pixels is "folded" back into a 2D grid of 64×64 dimensions. The batch matrix is reshaped from $(4096000, k)$ into a 4D tensor of size $(1000, 64, 64, k)$

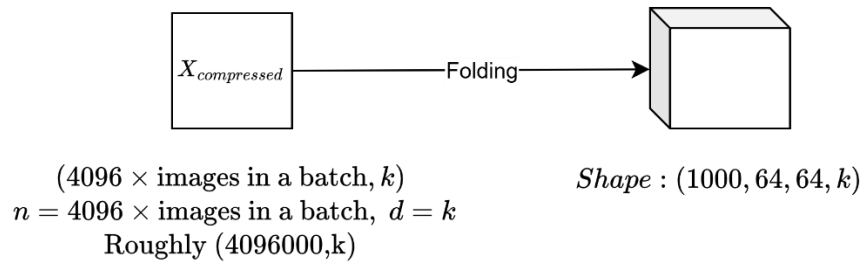


Fig 16. Folding

2.3.2 Image-Level Feature Flattening

To prepare the input for a traditional classifier like SVM, each reconstructed image cube must be represented as a single, unique feature vector.

Every $64 \times 64 \times k$ spatial-spectral cube is flattened into a 1D vector. The resulting feature vector has a length of $4096 \times k$. In fact, $k=5$ (achieving the 95% variance threshold), the feature vector length is 20,480. This is a massive reduction compared to the raw flattened size of 53,248, effectively eliminating redundant data while keeping the most discriminative information for the classifier.

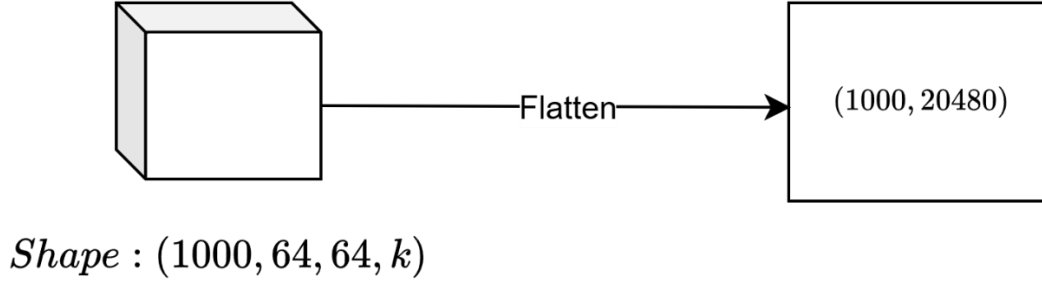


Fig 17. Flattening from folded

2.3.3 Global Matrix Consolidation

The final task of the pipeline is to stack all 27 independent batches into a single global training matrix. The processed feature vectors from all 27 batches are vertically concatenated. The pipeline concludes with a global feature matrix of size (27000, 20480) and a corresponding label vector of size (27000, 1)

$$X_{Final} = \text{stack}(X_{Flat1}, X_{Flat2}, \dots, X_{Flat27}) = \begin{bmatrix} X_{Flat1} \\ X_{Flat2} \\ \vdots \\ X_{Flat27} \end{bmatrix}$$

Fig 18. Final matrix – ready for SVM

Operation	Input Shape	Output Shape	Objective
Spatial Folding	(4,096,000, k)	(1000, 64, 64, k)	Restore pixel-to-pixel spatial relationships.
Image Flattening	(1000, 64, 64, k)	(1000, 4096×5)	Create a single vector per image.
Global Stacking	27 Processed Batches	(27000,20480)	Consolidate the full dataset for SVM.

Table 2. Summary of Data Transformations in Stage 3

CHAPTER 3: IMPLEMENTATION

This chapter details the technical execution of the pipeline, focusing on the software architecture, the custom-built dimensionality reduction engine, and the training of the Support Vector Machine (SVM) on the Kaggle platform.

3.1. Development Environment and Software Stack

The project was developed in a cloud-based environment to leverage high-performance computing resources.

Platform: Kaggle Notebooks.

Hardware Monitoring: We utilized the `tracemalloc` library to monitor peak RAM usage and the `time` library to track computational latency.

Libraries:

- NumPy & Pandas: For matrix manipulation and data handling.
- Scikit-learn: For implementing `train_test_split`, `SVC`, and performance metrics.
- Joblib: For persistent model serialization.

3.2. Implementation of the PCA Engine

While standard libraries offer PCA, we implemented a custom `PCAEngine` class to provide granular control over the Eigen-decomposition process and ensure memory efficiency during the transformation of 27 batches.

The engine calculates the `mean_vector` and `std_vector` from the first batch to establish a global scaling baseline. It solves the Characteristic Equation of the covariance matrix to derive eigenvalues and eigenvectors.

The engine automatically determines the number of principal components (k) by identifying the index where the cumulative variance ratio first exceeds 95%.

3.3. Training Workflow and Data Partitioning

After transforming all batches, the pipeline consolidates the data into a final training matrix stored in `.npy` format.

The resulting training matrix `X_TRAIN_FINAL` reached a shape of (27000, 20480). This confirms that with 95% variance, each 64×64 image was compressed into a feature vector of length 20,480. We employed a 80/20 Stratified Split:

- Training Set: 21,600 images.
- Testing Set (Blind Test): 5,400 images.

Ensuring the 10 terrain classes are distributed proportionally between the training and testing sets to prevent class imbalance bias.

3.4. Hyperparameter Optimization and Kernel Selection

Prior to the final training session, a rigorous optimization phase was conducted to identify the most effective configuration for the SVM classifier. Given the computational intensity of searching through a 27,000-image dataset, we strategically utilized a representative subset of **3,000 images**.

We first employed a **Grid Search** strategy to perform a comparative analysis between the Linear and Radial Basis Function (RBF) kernels. The results, as illustrated in the experimental logs, indicated a clear superiority of the **RBF kernel**, which achieved an accuracy of 62.96% compared to the 47.23% observed with the Linear kernel.

Kernel	Regularization (C)	Avg. Time (s)	Accuracy
Linear	1.0	11.87	47.23%
Linear	10.0	11.65	47.23%
RBF	1.0	10.70	62.62%
RBF (Best)	10.0	11.51	62.96%

Table 3. Kernel Comparison via Grid Search

To adjust the C parameter, we performed a Random Search with 15 trials (5 candidates x 3 CVs) on 3,000 sample images. Using a loguniform distribution(1, 100) helps to scan the entire value objectively.

Trial	C Parameter Value	Average Fit Time (s)	Cross-Validation Accuracy
1	5.6115 (Best)	8.14	63.08%
2	2.0513	8.77	62.87%
3	15.7513	10.01	62.38%
4	29.1063	10.02	62.25%
5	79.6945	8.70	61.63%

Table 4. C comparison via Random search

CHAPTER 4: RESULTS AND ANALYSIS

This chapter presents a comparative analysis between the raw multispectral data and the PCA-compressed features. The evaluation focuses on three key metrics: classification accuracy, training time, and memory consumption (RAM).

4.1. Experimental Setup and Data Constraints

To evaluate the scalability of the SVM model, we prepared datasets at three different scales: 3,000, 18,000, and 27,000 images. A critical discovery was made regarding the “raw” dataset (53,248 features per image).

At 18,000 images, the raw feature matrix occupied over 28GB of RAM, reaching the upper threshold of the Kaggle environment (30GB). Consequently, it was impossible to process the full 27,000 images in their raw state, as the memory overhead would lead to a system crash. This limitation underscores the necessity of the PCA-based pipeline developed in this research.

4.2. Comparative Performance Analysis

This section focuses on analysis three key metrics: accuracy, training time and memory consumption (RAM).

4.2.1. Accuracy Comparison

Accuracy reflects the model's ability to correctly identify terrain types. As shown in the data, PCA-compressed features consistently maintain high competitive accuracy compared to the high-dimensional raw data.

Sample Size	RAW Accuracy	PCA Accuracy
3K	74.11%	76.32%
18K	82.95%	81.36%
27K	ERROR	83.56%

Table 5. Accuracy comparison

At the 3K scale, PCA actually outperformed the raw data, likely due to the noise reduction effect of the PCA engine. While the raw data showed a slight edge at 18K, the margin is negligible compared to the resource savings achieved. The peak performance of **83.56%** was only attainable using the PCA pipeline at the full 27K scale.

4.2.2. Time Consumption (Computational Latency)

Time consumption is measured from the start of the training process to the completion of the blind test prediction.

Sample Size	RAW Time (s)	PCA Time (s)	Improvement
3K	179.1s	71.66s	~60% faster
18K	4081.8s	1555.2s	~62% faster
27K	ERROR	3884.69s	N/A

Table 6. Time comparison

The PCA pipeline demonstrated a dramatic reduction in temporal cost. At 18K images, the Raw training took over 1 hour, whereas PCA completed the task in approximately 26 minutes. This efficiency is a direct result of reducing the feature space from 53,248 to 20,480.

4.2.3. Memory Utilization (RAM Cost)

RAM usage is the most critical constraint in this research. The data below illustrates the peak memory footprint captured during execution.

Sample Size	RAW RAM (MB)	PCA RAM (MB)
3K	3,603.64	1,227.02
18K	28,852.82	7,323.64
27K	ERROR	10,969.23

Table 7. RAM comparison

At 18K images, the Raw dataset pushed the system to its absolute limit (28.8 GB), leaving almost no overhead for the operating system or kernel computations. In contrast,

PCA utilized only 7.3 GB for the same sample size, representing a 75% improvement in memory efficiency.

4.3. Comparison with Baseline Studies

To validate the competitiveness of our proposed PCA-SVM pipeline, we benchmarked our results against the original EuroSAT study conducted by Helber Patrick et al.[1] The comparison focuses on the 80/20 train-test split, which is the standard ratio used in their literature for high-performance evaluation.

Method	10/90	20/80	30/70	40/60	50/50	60/40	70/30	80/20	90/10
BoVW (SVM, SIFT, k = 10)	54.54	56.13	56.77	57.06	57.22	57.47	57.71	58.55	58.44
BoVW (SVM, SIFT, k = 100)	63.07	64.80	65.50	66.16	66.25	66.34	66.50	67.22	66.18
BoVW (SVM, SIFT, k = 500)	65.62	67.26	68.01	68.52	68.61	68.74	69.07	70.05	69.54
CNN (two layers)	75.88	79.84	81.29	83.04	84.48	85.77	87.24	87.96	88.66
ResNet-50	75.06	88.53	93.75	94.01	94.45	95.26	95.32	96.43	96.37
GoogLeNet	77.37	90.97	90.57	91.62	94.96	95.54	95.70	96.02	96.17

Fig 19. Accuracy in Helber Patrick et al. research

Method	Feature Extraction Method	Accuracy (%)
BoVW (SVM, SIFT, k=10)	Local Geometric Features	58.55%
BoVW (SVM, SIFT, k=100)	Local Geometric Features	67.22%
BoVW (SVM, SIFT, k=500)	Local Geometric Features	70.05%
Proposed PCA-SVM	Global Spectral Features (PCA)	83.56%

Table 8. Comparative Analysis with Original EuroSAT Baseline (80/20 Split)

The results indicate that our PCA-SVM approach outperforms the best SIFT-based configuration by **13.51%**. This suggests that for multispectral satellite imagery, the global spectral information (retained through PCA) is far more discriminative than local texture features like SIFT, which were originally designed for traditional RGB photography.

When compared to contemporary Deep Learning models introduced in the same study, our approach provides a compelling balance between accuracy and computational cost. While our model falls short of advanced architectures like ResNet-50 (96.43%) and GoogLeNet (96.02%), it remains competitive with simpler Deep Learning structures such

as the 2-layer CNN (87.96%). Unlike CNNs and Residual Networks that necessitate high-end Graphics Processing Units (GPUs) and extensive training hours, our PCA-SVM pipeline operates efficiently on standard CPUs.

4.4. Confusion Matrix Analysis

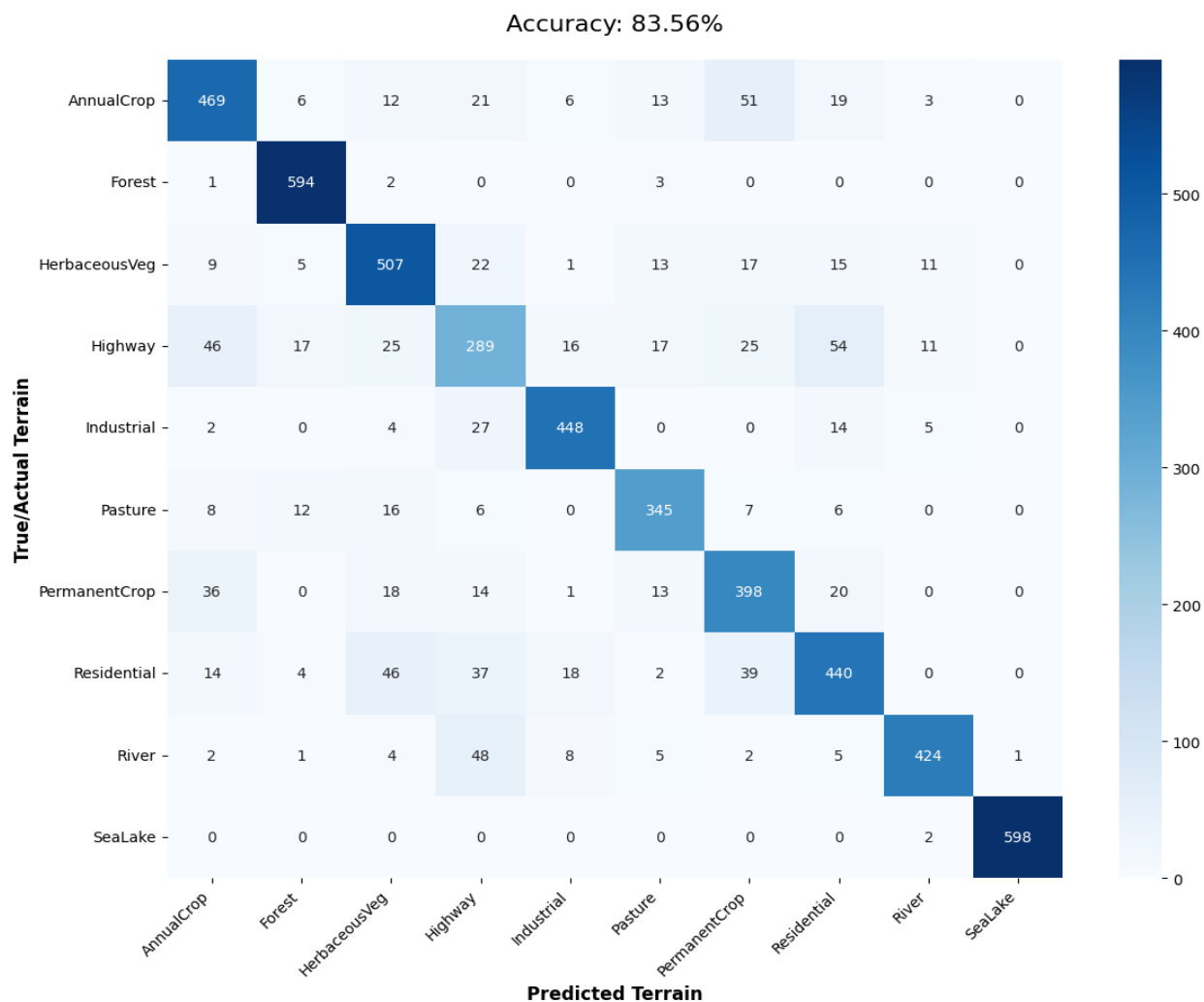


Fig 20. Confusion matrix

To gain a deeper understanding of the model's behavior across different land cover classes, we analyze the Confusion Matrix generated from the 83.56% accuracy run. The matrix reveals how spectral similarities between certain terrain types influence classification errors.

The model achieves near-perfect classification for SeaLake (598/600) and Forest (594/600). These classes possess distinct spectral signatures—deep water absorption and dense vegetation chlorophyll reflectance

The most significant confusion occurs between Highway and River (48 instances). From a multispectral perspective, both features often appear as linear structures with similar reflectance properties when asphalt and turbid water are involved.

There is a noticeable overlap between AnnualCrop, PermanentCrop, and Pasture. For instance, 51 instances of AnnualCrop were misclassified as PermanentCrop. This is expected as these classes share high NDVI values, making them spectrally adjacent during certain seasonal cycles.

4.5. Advantages and Limitations

****Advantages:***

Resource Efficiency: The primary advantage of the PCA-SVM pipeline is its minimal hardware requirement. By reducing the feature space by over 60%, we achieved a 75% reduction in RAM usage, enabling large-scale processing on standard cloud infrastructure without expensive GPU resources.

Noise Robustness: The PCA dimensionality reduction acts as a latent feature extractor. By discarding low-variance components, the model effectively ignores sensor noise and redundant spectral information, leading to higher accuracy at smaller sample sizes (76.32% vs 74.11% at 3K).

Hardware Accessibility: A major advantage of this pipeline is its ability to be deployed on mainstream hardware systems. While modern Deep Learning models require large amounts of VRAM from specialized GPUs (such as NVIDIA A100 or T4), the PCA-SVM model can run smoothly on the CPUs of personal computers or low-configuration cloud servers.

****Limitations:***

Spatial Context Loss: As a pixel-based or flattened-feature approach, the model focuses primarily on spectral signatures. It lacks the convolutional ability to recognize complex spatial patterns (e.g., the specific geometry of buildings), which explains the confusion between linear features like highways and rivers.

Sensitivity to Preprocessing: The performance of the SVM is highly dependent on the scaling and centering performed during the PCA phase. Any inconsistency in the initial data distribution across batches could lead to sub-optimal hyperplane separation.

Information Trade-off: Although PCA significantly reduces computational resources, retaining only 95% of the variance means that the remaining 5% of spectral information is discarded. In some specific cases, this 5% may contain important features for distinguishing highly similar classes (such as different crop types). This explains why the accuracy of PCA at 18K image resolution is slightly lower (1.59%) compared to using complete Raw data.

CONCLUSION

This research successfully developed and evaluated a high-efficiency classification pipeline for multispectral satellite imagery using the EuroSAT dataset. By integrating a custom PCA engine with a Support Vector Machine (SVM) classifier, we addressed the fundamental challenge of high-dimensional data processing in Earth observation. The implementation on the Kaggle platform demonstrated that dimensionality reduction is not only an optimization tool but a prerequisite for large-scale analysis on standard cloud infrastructure.

In conclusion, the combination of Singular Value Decomposition and Support Vector Machines offers a powerful, efficient, and reliable framework for land cover classification. By prioritizing the balance between computational cost and predictive precision, this research provides a viable pathway for large-scale satellite data analysis in resource-constrained environments.

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APPENDIX

Link to Kaggle for code resource:

1. PCA/SVM Training: <https://www.kaggle.com/code/riatleozia/eurosat-image-classifier-with-svm-after-pca>
2. Data processing: <https://www.kaggle.com/code/riatleozia/ingestion-for-pca-svm>
3. Raw data SVM Training: <https://www.kaggle.com/code/riatleozia/eurosat-image-classifier-with-svm-raw>